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Care Management to Promote Treatment Adherence in Patients with Cognitive Impairment and Vascular Risk Factors: A Demonstration Project

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Abstract

Dementia prevention is highly important. Improved control of vascular risk factors has the potential to decrease dementia risk, but may be difficult. Therefore, we developed and piloted a care management protocol for Veterans at risk for dementia. We enrolled 32 Veterans with diabetes and hypertension, at least one of which was poorly controlled, and cognitive impairment. Participants were randomly assigned to a 6-month care management intervention or to usual care. At enrollment, 6-months and 12-months, we assessed cognitive performance, mood, and diabetes and hypertension control. At follow-up, diastolic blood pressure was lower in intervention participants at 6 months (p=.041) and 12 months (p=.022); hemoglobin A1c, global mental status and mood did not differ between groups. Recall of a distractor list (p=.006) and logical memory long-delay recall (p=.036) were better at 6 months in the intervention group (p=.006). Care management may contribute to improved control of dementia risk factors.

Key words: Dementia, prevention, chronic disease, care management, treatment adherence, type 2 diabetes, hypertension.

Introduction

Neurodegenerative disorders, including Alzheimer's disease and vascular dementia, cause significant disability among older adults, with negative effects on quality of life for patients and families, and financial costs to society. Unfortunately, limited treatment options exist once dementia has been diagnosed. Prevention of these disorders is therefore highly important. Recent research supports the role of vascular risk factors, including hypertension and diabetes, as potentially modifiable contributors to neurodegenerative disease (1).

Adherence to treatment regimens among patients with multiple chronic conditions is often suboptimal, and adherence may be even more difficult for patients with cognitive deficits (2). Therefore, patients with comorbid diabetes and hypertension who have cognitive deficits are at high risk of worsened control of their vascular risk factors and thus, worsened cognitive status. Such patients, while at high risk of dementia, may benefit significantly from improved control of chronic conditions.

Care management has been used to improve adherence among patients with various chronic conditions (hypertension, diabetes, depression, etc. (3)). Little is known, however, about how to adapt care management protocols to patients with cognitive deficits, despite the high prevalence of cognitive disorders in an aging patient population. Further, although care management is intended to prevent complications of chronic illness, its potential to prevent cognitive decline through improved control of risk factors has not been studied. Our goal in this demonstration project was to provide care management, specifically tailored to Veterans with existing cognitive impairment to improve disease control and prevent or slow further cognitive decline.

Methods

Recruitment

Veterans were eligible for recruitment if they were older than 54 years and had both diabetes and hypertension, with poor control of at least one disorder. Poor control was defined as systolic blood pressure > 140 mm Hg or hemoglobin A1c > 7.0 %. Eligibility for the study also required a cognitive deficit as determined by study neuropsychological assessment procedures or pre-existing memory disorder diagnosis (including Mild Cognitive Impairment (MCI), Alzheimer's disease, vascular dementia, or Cognitive Disorder Not Otherwise Specified). Note that study enrollment was completed prior to revisions to the Diagnostic and Statistical Manual, 5th edition (4). Therefore, the terms "dementia" and "Mild Cognitive Impairment" are used throughout this report. Participants with dementia were eligible if their deficits were mild enough to allow comprehension of concerns raised by the care manager and ability to

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make and act on decisions about health-related behaviors such as exercise and medications. Participants with dementia were also required to have a caregiver who enrolled in the study. The optimal age of intervention for dementia prevention is still under investigation, but neurodegenerative changes are thought to begin in the brain a decade prior to cognitive symptoms (5). Individuals aged 54 years and older may already be experiencing the neurodegenerative changes in the brain that precede a diagnosis of dementia, but the neurodegenerative process may be still be amenable to intervention.

In-person assessment

Multiple recruitment strategies were employed in order to recruit from the broadest possible population of VA patients. Recruitment strategies included: in-clinic recruitment, mailed letters followed up by in-person or telephone screening, and referrals from VA clinicians.

In-person and mail recruitment

VA patient names were obtained from VA databases and screened for medical eligibility. Eligible Veterans were then recruited in-person from VA primary care clinics, or were recruited by mail. In the mail recruitment process, a letter describing the study was mailed to eligible patients and followed up with a phone call.

For both the mail and in-clinic recruitment processes, a two-step screening process was used to determine cognitive eligibility. Patients first completed a brief cognitive screening. Those whose screening results suggested cognitive impairment were then invited to participate in the study. These participants completed a one-hour cognitive screening including assessment of mental status, attention, verbal memory, visual memory, verbal fluency, functional abilities and mood.

Telephone-only assessment

Travel to the VA for screening and cognitive assessment was a barrier to participation for some Veterans. Therefore, we offered a telephone screening and assessment process for interested Veterans.

Referrals

Potential participants could also be referred by VA clinicians. Veterans referred from Memory Disorders Clinic had already completed a clinical memory assessment; memory test data from the clinical assessment were used in lieu of research screening to determine eligibility.

Instruments

In-person screening

The initial screening included brief instruments (Mini-Cog (6)) or Montreal Cognitive Assessment (MoCA; (7)). A score of 4 or 5 on the Mini-Cog was considered cognitively intact and therefore ineligible. A score of 3 was considered marginal; clinical judgment was required to determine eligibility. A score of <3 was considered eligible.

After the initial screening, participants provided informed consent and completed a one-hour memory assessment. The memory assessment included the Modified Mini Mental Status Examination (3-MSE), Wechsler Memory Scale, 3rd Edition, Logical Memory subtest (8); the California Verbal Learning Test, 2nd Edition (9); category and phonemic fluency, and the Geriatric Depression Scale (10).

Telephone assessment

The telephone assessment used the Rivermead story (immediate recall) subtest (11), category fluency, Telephone Interview for Cognitive Status (12) and Center for Epidemiologic Studies Depression scale (13).

Randomization

Participants were assigned to intervention or usual-care group using a pseudo-randomization procedure (selecting among 3 sealed envelopes, 1 of which contained a usual-care assignment and 2 of which contained an intervention assignment).

Care management

Veterans randomized into the intervention group worked with the RN care manager to develop individualized plans to address barriers to treatment adherence. The intervention primarily used telephone contact in order to minimize travel for participants, although some participants chose to have in-person visits with the care manager. The intervention was adapted to each patient's needs, but included education about disease management, discussion of barriers to adherence, and encouragement of patient engagement with other services such as social work, nutrition, and pharmacy. Spouses and other caregivers were encouraged to participate in conversations if permitted by the participant. The intervention continued for a 6-month period.

Table 1. Baseline Demographics

(N=32)	Usual-Care (%)	Intervention (%)	Total (%)	p
Male	16 (100)	15 (94)	31 (97)	.5
Age (years)	65.23+4.6	65.1+4.7	65.2+4.6	.933
Ethnicity:				*.475*
White	11 (68.8)	11 (68.8)	22 (68.8)	
African-American	3 (18.8)	5 (33.3)	8 (25)	
Cognitive diagnosis				*.567*
Mild Cognitive Impairment (MCI)	9 (56.3)	7 (43.8)	16 (50)	
Alzheimer's disease	1 (6.3)	0 (0)	1 (3.1)	
Other cognitive deficits	6 (37.5)	9 (56.3)	15 (46.9)	
Mental health diagnoses:				
Depression	5 (31.3)	8 (50)	13 (40.1)	.384
PTSD	8 (50)	5 (31.3)	13 (40.1)	.275
Other psychiatric	2 (12.5)	4 (25)	6 (18.8)	.426
Sleep apnea	9 (56.3)	6	15	.118
Body Mass Index	32.6 + 5.3	32.8 + 5.8	.661	
Systolic blood pressure (mm Hg)	127.9 + 13.6	133.8 + 16.2	131 + 15	.289
Diastolic blood pressure (mm Hg)	69.7 + 10.3	76 + 11	73 + 11	.110
Hemoglobin A1c (%)	7.7 + 1.7	7.5 + 1.3	7.6 + 1.4	.684
MMSE[1]	26.9 + 2.5	26.5 + 2.1	26.7 + 2.3	.721
GDS[2]	16.9 + 6.7	13.2 + 8.6	15 + 7.8	.247
Logical Memory I (raw)	18.3 + 6.1	24.5 + 5.5	21.6 + 6.4	.032*
Logical Memory II (raw)	17.2 + 7.7	20.8 + 6.0	19.1 + 6.9	.270
Verbal fluency (semantic)	15.2+5.5	16.5+4.9		.942
Verbal fluency (phonemic)	26.6+11.6	32+13		.615
CVLT[3] listb (raw)	5+2.2	3.8+1.5		.081
CVLT[3] short delay free recall	6.45+3.3	7.46+3.7		.640
CVLT[3] long delay free recall	6.75+5	6.92+3.2		.933

1: MMSE: Mini Mental State Examination; 2: GDS: Geriatric Depression Scale; 3: CVLT: California Verbal Learning Test, 2nd Edition

Data collection

Cognitive data

The one-hour memory assessment was repeated at 6 and at 12-months for participants who had completed this assessment at enrollment. Participants who had completed a telephone assessment at enrollment repeated the telephone assessment at 6 and 12 months.

Health status

Laboratory values and other health-related variables were abstracted from each Veteran's electronic health record at baseline, 6-months and 12-months. Variables included: demographic information, systolic and diastolic blood pressure, weight, hemoglobin A1c, and lipids, diagnoses of sleep apnea, cognitive disorders, mental health conditions, hyperlipidemia and medications.

Data analysis

Analyses of the dependent measures were conducted using mixed effects linear regression, with treatment and follow-up interval as fixed effects, and participant as a random effect. Restricted maximum likelihood and the Kenward-Roger approximation for degrees of freedom were used to address the small sample size. Disparate sets of respondents among the dependent measures precluded multivariate analyses. Analyses were not adjusted for multiple comparisons. We believe that this approach is appropriate given the exploratory nature of this study.

Table 2. This table shows the raw scores for each of the measures at 6 and 12 months by group

6-Month Results	Usual-care	Intervention	Test statistic	P
MMSE[1]	26.80±1.8	26.89±2.8	F=.007	.934
GDS[2]	16.8±7.7	10.0±5.2	t=-.92	.364
Diastolic blood pressure (mm Hg)	72.7±11.2	70.2±13	t=-2.09	.041*
Systolic blood pressure (mm Hg)	129+±10.9	128±14.2	t=-1.1	.274
Logical Memory II	17±7.63	23±5.8	t=2.19	.036*
Hemoglobin A1c	7.6±1.8	7.5±.96	Z=-.17	.864
CVLTb raw	3.64±1.3	5.22±1.8	t=2.94	.006*
12-month results				
MMSE1	28.09±1.2	26.67±2.4	F=2.97	.102
GDS2	14.4±5.9	9.0±8.9	t=-.28	.78
Diastolic blood pressure (mm Hg)	79.2±14.6	74.3±12.7	t=-2.35	.022*
Systolic blood pressure (mm Hg)	137±19.3	132±17.6	t=-1.95	.056
Logical Memory II	21.1±9.33	23±8.17	t=.94	.354
Hemoglobin A1c (%)	7.4±1.37	7.23±.98	Z=-.15	.883
CVLTb3 raw	4.64±.92	4.44±1.8	t=1.18	.246

p values < 0.05 are starred; 1: MMSE: Mini Mental State Examination; 2: GDS: Geriatric Depression Scale; 3: CVLT: California Verbal Learning Test, 2nd Edition

Statistical analyses were conducted using Statistical Package for the Social Sciences (14) and Stata (15). All study procedures were approved by the VA Puget Sound Institutional Review Board.

Results

Screening/Enrollment

Overall, 136 veterans were assessed for eligibility for this study. Of those assessed, 87 were recruited as described in the methods and 49 were referred by Memory Disorders Clinic providers after being diagnosed with cognitive impairment. Of these 136 individuals, 58 did not meet inclusion criteria, 6 refused to participate, and 39 were excluded for other reasons. 33 veterans were randomized. One participant in the usual-care group was excluded after randomization due to determination of no cognitive impairment, leaving a combined total of 32 Veterans across all referral sources. Of enrolled participants, 5 were diagnosed with mild dementia, the remainder with mild cognitive deficits.

Demographics

The majority of participants were male. One-fourth were African-American. The mean GDS score was 16.9 ± 6.68 for usual care and 13.2 ± 8.64 for intervention participants; these scores are consistent with the presence of depression. Descriptive statistics are presented in Table 1. The intervention group had a higher score on logical memory, otherwise baseline demographics did not differ between groups.

In unadjusted analyses, diastolic blood pressure was lower in intervention than usual-care participants at 6 months (z=-2.17, p=.030) and at 12 months (z=-2.44, p=.015). Systolic blood pressure was lower in intervention participants at 12 months (z=-2.02, p=.043) but not at 6 months (z=-1.15, p=.252). Hemoglobin A1c (6-month: z=-.18, p=.856; 12-month: z=-.16, p=.876) did not differ between groups. Global cognitive status did not differ between groups. Immediate and delayed recall for a list of unrelated words did not differ between groups, except that recall of a distractor list was better at 6 months in the intervention group (z=3.14; p=.002). Logical memory long-delay recall was better at 6-months in intervention than usual-care participants (z=2.35, p=.019). Depression scores did not differ between groups. Results are presented in Table 2.

Conclusions/Discussion

This study demonstrates the potential of care management approaches to engage patients with cognitive limitations, with the ultimate goal of dementia prevention. Care management was individually adapted to the specific needs of each participant (nutrition education, engagement of family caregivers, and so forth). Although global cognitive status did not differ between intervention and usual-care groups, recall of short stories and recall of a word list presented once were superior in the intervention group at 6-months. Diastolic blood pressure control was superior in the intervention group at 6-months and both systolic and diastolic blood

pressure control were superior at 12-months. These results suggest that care management designed for Veterans with cognitive impairments and risk factors for dementia can positively affect both cognitive and vascular health related variables. Although blood pressure improvements were sustained at 12 months, cognitive improvements were not; these results suggest that follow-up care may be important to sustain cognitive benefits.

The presence of elevated GDS scores among our participants suggests that depression is a common co-morbid condition among veterans with cognitive impairment, diabetes and hypertension, and may be an important target for future intervention.

The high enrollment of African American Veterans is interesting, especially given historical challenges to research recruitment in this population. Further research is needed to understand how best to serve the needs of African American older adults with cognitive deficits and vascular risk factors.

To our knowledge, this is the first protocol developed to prevent dementia in patients with vascular risk factors who are beginning to experience mild cognitive symptoms. Given the prevalence and impact of dementia, further exploration of this promising avenue of prevention is warranted.

Limitations

This study was a pilot study with a small sample size. More research is needed to determine the optimal care management protocol for Veterans, and older adults in general, who have multiple vascular risk factors for dementia. This study was conducted among Veterans receiving care in VA; more research is needed to determine how to serve the needs of patients with cognitive impairments in other settings.

We did not conduct a formal measure of engagement in the intervention. Anecdotally, we found that engagement varied, with some participants seeking little contact with the care manager and others frequently calling the care manager and acting on recommendations. However, this pattern of variable engagement is likely to reflect actual implementation of care management in the clinical setting.

We did not examine changes in medications or other treatments during study participation. It is possible that medication changes were initiated by physicians due to clinical changes independent of study participation. Medication changes could also have been affected by study participation (for example, a patient might have been encouraged by the care manager to discuss side effects with his or her physician). This topic is complex and requires further study.

Due to the limited scope of this project, we abstracted laboratory data from the VA electronic medical record rather than obtaining direct measurements. Therefore laboratory values are not available for each participant at each time point.

Despite these limitations, this study demonstrates the potential benefits of care management for older adults with cognitive deficits and vascular risk factors, and suggests a useful avenue for further dementia prevention research. Specifically, this study demonstrates that improvement in important disease markers (systolic and diastolic blood pressure, logical memory and list-learning) may be achieved in the absence of improved global mental status, and suggests the importance of mood as a target of intervention for older adults with chronic illness and cognitive impairment.

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Conflict of Interest: There are no conflicts of interest for any of the authors.

Ethical standards: Participants provided written informed consent to participate in this study. This consent procedure and all study procedures were approved by the VA Puget Sound IRB.

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